



Single/2~24 input RS485 output instruction manual

1. Physical interface

- Serial communication uses RS485.
- The transmission mode is half-duplex asynchronous mode, with 1 start bit, 8 data bits, and 1 stop bit, no checksum.
- The data transmission rate defaults to 9600bps (can be modified). Using MODBUS communication protocol RTU mode.
- The factory default address is 0x01, and the default baud rate is 9600bps.
- All register address bytes, register number bytes, and data bytes in the command are high-order first. The low-order byte comes first; the low-order byte of the CRC check code comes first, and the high-order byte comes last.
- The CRC check code is calculated by the sending or receiving device through the CRC algorithm. The test software is generally it will have its own automatic CRC calculation function. Different device commands have different CRC check codes and must not be used randomly enter.

2. Read all data command example

2.1 Single input

- Single-channel COMMAND (single-channel input and download command)

slave device address	function code	Starting register address		Number of registers		CRC-L	CRC-H
01H	03H	00H	56H	00H	01H	64H	1AH

- Single-channel RETURN (single-channel input return information)

slave device address	function code	Number of bytes in data area	Single channel data		CRC-L	CRC-H
			A/D-H	A/D-L		
01H	03H	02H	24H	F6H	23H	02H

2.2 Dual input

- Single-channel COMMAND (single-channel input and download command)

slave device address	function code	Starting register address		Number of registers		CRC-L	CRC-H
01H	03H	00H	10H	00H	02H	C5H	CEH

- Single-channel RETURN (single-channel input return information)

slave device address	function code	Number of registers	1st channel output		2cnd channel output		CRC-L	CRC-H
			A/D-H	A/D-L	A/D-H	A/D-L		
01H	03H	04H	1FH	3FH	0AH	74H	CBH	6CH

Note: Special note on the combination of current and voltage dual-channel inputs corresponding to RS485 outputs:
RS485 first channel output: voltage; RS485 second channel output: current.

2.3 Three-phase (three-channel) inputs

- Single-channel COMMAND (single-channel input and download command)

slave device address	function code	Starting register address		Number of registers		CRC-L	CRC-H
01H	03H	00H	10H	00H	03H	04H	0EH

- RETURN (return information)

slave device address	function code	Number of registers	1st channel output		2nd channel output		3rd channel output		CRC-L	CRC-H
			A/D-H	A/D-L	A/D-H	A/D-L	A/D-H	A/D-L		
01H	03H	06H	1EH	AFH	1EH	1CH	1EH	B1H	CBH	6CH

- Special instructions for three-phase voltage input:

- Three-phase three-wire voltage input corresponds to RS485 output: RS485 first channel output: UAB
RS485 second channel output: UBC
RS485 third channel output: UAC
- Three-phase four-wire voltage input corresponds to RS485 output: RS485 first channel output: UAN
RS485 second channel output: UBN
RS485 third channel output: UCN
- Three-phase current input corresponds to RS485 output: RS485 first channel output: IA
RS485 second channel output: IB
RS485 third channel output: IC

2.4 Multiple inputs (4~24 channels)

- Read 4~24 channels of data COMMAND (read 4~24 channels of data and download the command)

slave device address	function code	Starting register address		Number of registers		CRC-L	CRC-H
01H	03H	00H	10H	00H	XXH	XXH	XXH

Note: 1. In "Number of Registers": XX=number of sensor measurement channels, for example, 4-channel XX=04, 8-channel XX=08, 12-channel XX=0C, 18-channel XX=12, 24-channel XX=18.

- Starting from register 10, read XX register data continuously. Each measured value data occupies one register.

- Read data RETURN (read return information)

slave device address	function code	Number of bytes in data area	1st channel data		2nd~23rd channel data	24th channel data		CRC-L	CRC-H
			A/D-H	A/D-L		A/D-H	A/D-L		
01H	03H	XXH	1EH	AFH		1EH	B1H	XXH	XXH

- Note:** 1. In "Number of Bytes in Data Area", $XX=2 \times \text{Number of Measurement Channels}$
 2. The XX in the CRC check code should be derived based on actual data
 3. Return data restoration instructions example:
 3.1 A/D-H = the high 8 bits of the measured data, A/D-L = the low 8 bits of the measured data.
 For example, in the single-channel return data in 2.1 above, A/D-H=24H and A/D-L=F6H are synthesized into hexadecimal =24F6.

3.2 Return data calculation formula:

Basic calculation method: Calculate according to the linear relationship between the input rated value range and the output data 0~10000 digital quantity.

$$Bx = \text{DATA} \times (\text{Af} - \text{A0}) / 10000 - |\text{A0}|$$

Bx --Measurement value of the calibrated sensor

DATA--returned data value (A/D-H+D/L converted to decimal value)

Af --the upper limit of the input nominal value of the calibrated sensor

A0 --The lower limit of the nominal input value of the calibrated sensor

For example (1): The nominal value of the sensor input is the voltage 0~700V (effective value),

The above 2.1 single-channel return data DATA=24F6 (hexadecimal) =9462 (decimal),

Then the measured actual voltage

$$Bx = \text{DATA} \times (\text{Af} - \text{A0}) / 10000 - |\text{A0}|$$

$$= 9462 \times (700 - 0) / 10000 - |0| = 662.34\text{V}$$

For example (2): The nominal value of the sensor input is the bidirectional DC voltage -100V~+200V,

The above 2.1 single-channel return data DATA=24F6 (hexadecimal) =9462 (decimal),

Then the measured actual voltage

$$Bx = \text{DATA} \times (\text{Af} - \text{A0}) / 10000 - |\text{A0}|$$

$$= 9462 \times (200 - (-100)) / 10000 - |-100| = 183.86\text{V}$$

Note: 1. The current sensor is calculated in the same way as the above voltage sensor.

2. The data type is unsigned; data ending with "H" only represents that data. It is in hexadecimal, please do not enter "H" during communication detection.

3. Example of command to modify address

● COMMAND (download command, address changes from ADD1-01 to ADD2-02)

slave device address	function code	Starting register address		Number of registers		Number of data bytes		Data written to register		CRC-L	CRC-H
01H (ADD1)	10H	00H	57H	00H	01H	02H		00H	02H (ADD2)	2AH	76H

● RETURN (return information)

02H	10H	00H	57H	00H	01H	B0H	2AH
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Note: 1. The above command ADD1 is the address number of the original sensor, and ADD2 is the address number that is about to be changed.

2. The factory default address of the sensor is 01H.ADD1, and the settable range of ADD2 is 00-FF, with a total of 256 addresses.

3. If you don't know what the current address is, you can use the FA command to reset the address to 01 (factory address). The specific download command is as follows (COMMAND): FAH 10H 00H 57H 00H 01H 02H 00H 01H 1DH 43H

4. When ADD1=FAH, it is a broadcast command. No matter what the sensor address is on the RS485 bus, this command will always change the sensor address to ADD2.

4. Example of command to modify the baud rate (change the baud rate to 4800bps)

● COMMAND (download command)

slave device address	function code	Starting register address		Number of registers		Number of data bytes		Data written to register		CRC-L	CRC-H
01H	10H	00H	20H	00H	01H	02H	00H	05H (Changed baud rate code)		61H	33H

Note: Baud rate code setting

03-1200bps ; 04-2400bps ; 05-4800bps

06-9600bps ; 07-19200bps;

After writing the above baud rate code into the register, follow the baud rate corresponding to the code just communicate.

5. Message format

5.1 Function code 03H-Query slave register data

● Host message (download)		● Slave message (response)	
slave address	1 byte (00H~FFH)	slave address	1 byte (00H~FFH)
function code	1 byte (03H)	function code	1 byte (03H)
Starting register address	2 bytes	Number of bytes in data area	2*Number of registers 1 byte
Number of registers	2 bytes	data area	Number of bytes = 2X number of registers
CRC check code	2 bytes	CRC check code	2 bytes

5.2 Function code 10H-Set the slave

● Host message (download)		● Slave message (response)	
slave address	1 byte (00H~FFH)	slave address	1 byte (data written to register)
function code	1 byte (10H)	function code	1 byte (10H)
Register address	2 bytes (00H 57H or 00H 20H)	Register address	2 bytes

Number of registers	2 bytes	Number of registers	2 bytes
Number of bytes in data area	1 byte (2X number of registers)	CRC check code	2 bytes
Data written to register	Number of bytes = 2X number of registers		
CRC check code	2 bytes		

● **Noted:** CRC check code low bit first, high bit last; register address, register number, the data are all in high order first and low order last.

6. Register description

6.1 Single input

Register address	Register contents	Number of registers	Register status	data range
0056H	Single channel power	1	read only	0~10000

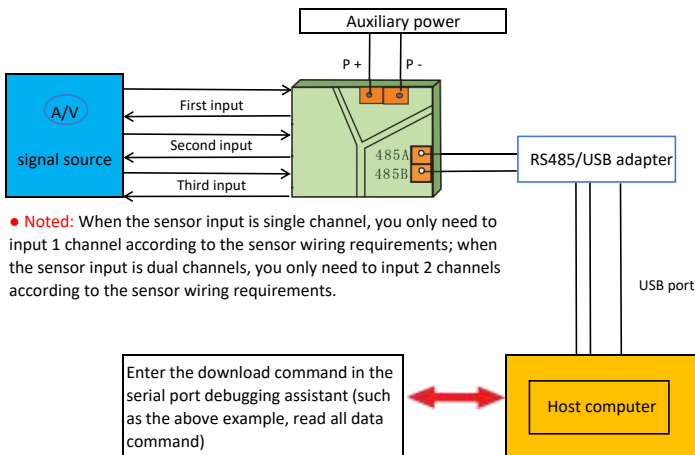
6.2 Multiple inputs (2~24 channels)

Register address	Register contents	Number of registers	Register status	Rated range	maximum range
0010H	2st Channel Power	1	read only	0~10000	0~12000
0011H	2nd Channel Power	1	read only	0~10000	0~12000
0012H	3rd Channel Power	1	read only	0~10000	0~12000
0013H	4th Channel Power	1	read only	0~10000	0~12000
0014H	5th Channel Power	1	read only	0~10000	0~12000
0015H	6th Channel Power	1	read only	0~10000	0~12000
0016H	7th Channel Power	1	read only	0~10000	0~12000
0017H	8th Channel Power	1	read only	0~10000	0~12000
0018H	9th Channel Power	1	read only	0~10000	0~12000
0019H	10th Channel Power	1	read only	0~10000	0~12000
001AH	11th Channel Power	1	read only	0~10000	0~12000
001BH	12th Channel Power	1	read only	0~10000	0~12000
001CH	13th Channel Power	1	read only	0~10000	0~12000
001DH	14th Channel Power	1	read only	0~10000	0~12000
001EH	15th Channel Power	1	read only	0~10000	0~12000
001FH	16th Channel Power	1	read only	0~10000	0~12000
0020H	17th Channel Power	1	read only	0~10000	0~12000
0021H	18th Channel Power	1	read only	0~10000	0~12000
0022H	119th Channel Power	1	read only	0~10000	0~12000
0023H	20th Channel Power	1	read only	0~10000	0~12000
0024H	21st Channel Power	1	read only	0~10000	0~12000
0025H	22nd Channel Power	1	read only	0~10000	0~12000
0026H	23rd Channel Power	1	read only	0~10000	0~12000
0027H	24th Channel Power	1	read only	0~10000	0~12000

● **Noted:** No. 1~24 power supply" corresponds to the "No. 1~24" input parameters of the sensor label.

7. Schematic diagram of sensor and computer connection

● The sensor can be connected to the computer through RS485/RS232 or RS485/USB converter, and communicate with it using the serial port debugging assistant to check whether the sensor is normal.



● **Noted:** 1. Sensor indicator light status, if the red light interval flashing frequency is 500ms, it is working normally;

The blue light flashes during normal communication and goes out when there is no communication; please connect the sensor correctly. The output terminals A and B of the sensor are reversely connected and cannot communicate; the sensor has been accurately calibrated when leaving the factory. Users generally do not need to verify again. (Some models have no indicator light, just use it normally)

2. Sampling frequency >200ms

8. The sensor adopts the RTU mode in the MODBUS communication protocol

RTU frame

Using RTU mode, message sending must start with a pause interval of at least 3.5 character times. This is easiest to implement with various character times at network baud rates (as shown in T 1-T2-T3-T4 in the figure below). The first field transferred is the device address. The transmission characters that can be used are hexadecimal 0...9,A...F. Network devices continuously monitor the network bus, including during pause intervals. When the first field (address field) is received, each device decodes it to see if it is destined for its own. After the last transmitted character, a pause of at least 3.5 characters marks the end of the message. A new message can be started after this pause.

The entire message frame must be transmitted as one continuous stream. If there is a pause of more than 1.5 characters before the frame is completed, the receiving device will flush the incomplete message and assume that the next byte is the address field of a new message. Likewise, if a new message begins after the previous message in less than 3.5 characters, the receiving device will consider it a continuation of the previous message. This will cause an error because the value in the last CRC field cannot be correct. A typical message frame looks like this:

start bit	Device address	function code	data	CRC	Check terminator
T1-T2-T3-T4	8Bit	8Bit	n*8Bits	16Bit	T1-T2-T3-T4

■9.Sensor input and output table (theoretical value)

input value percentage(%)	input value	Rs485 output	decimal
0%	A0	0x0000	0
20%	$A0 + (Af - A0) * 20\%$	0x07D0	2000
40%	$A0 + (Af - A0) * 40\%$	0x0FA0	4000
60%	$A0 + (Af - A0) * 60\%$	0x1770	6000
80%	$A0 + (Af - A0) * 80\%$	0x1F40	8000
100%	Af	0x2710	10000

Af --the input nominal value of the calibrated sensor

A0 --Lower limit of nominal input value of the calibrated sensor

Appendix 1: MOD BUS_CRC 16 check code calculation method

The cyclic redundancy check CRC area is 2 bytes and contains a 16-bit binary data. The sending device calculates the CRC value and attaches the calculated value to the message. When the receiving device receives the message, it recalculates the CRC value and compares the calculated value with the received time value in the CRC area. If the two are not the same, an error occurs.

At the beginning of CRC computation, initialize the 16-bit register with all bits set to "1". Then, place the data of two adjacent 8-bit bytes into the current register. Only the 8 bits of data from each character are used for CRC generation, excluding the start bit, stop bit, and parity bit, which are not included in the CRC.

During CRC generation, perform an XOR operation between each 8-bit data and the value in the register. Shift the result one position to the right (towards LSB), filling the MSB (most significant bit) with "0". Check the LSB (least significant bit), if LSB is "1", XOR it with a predetermined fixed value; if LSB is "0", no XOR operation is performed.

Repeat the aforementioned process until shifting has been performed 8 times. After completing the 8th shift, XOR the next 8-bit data with the current value in the register. After processing all the information, the final value in the register represents the CRC value.

The process of generating CRC:

- 1.Set the 16-bit CRC register to FFFFH.
- 2.Perform an XOR operation between the lower 8 bits of the CRC register and the first 8-bit data, and place the result into the CRC register.
- 3.Shift the CRC register one position to the right, fill the MSB with zero, and check the LSB.
- 4.(If LSB is 0): Repeat step 3, then shift one position to the right.
- (If LSB is 1:) Perform an XOR operation between the CRC register and A001H.
- 5.Repeat steps 3 and 4 until 8 shifts are completed, completing the processing of an 8-bit byte.
- 6.Repeat steps 2 through 5, processing the next 8-bit data, until all bytes are processed.
7. The final value in the CRC register represents the CRC value.
8. When inserting the CRC value into the information, the high 8 bits and low 8 bits should be placed separately.

Insert the CRC value into the information.

When sending the 16-bit CRC value in the information, transmit the low 8 bits first, followed by the high 8 bits. If the CRC value is 1A64:

slave device address	function code	Starting register address	Number of registers	CRC-L	CRC-H
				64H	1AH